Neural Networks & Backpropagation

The Perceptron Algorithm

Binary classification is the task of classifying data into two categories. Given data with labels, we seek to learn the classifier. Inspired by neurons in the brain, the Perceptron algorithm learns a binary classifier from labeled data.

Problem Setting

Consider labeled data $(x, l(x))$, where $x \in X \subset \mathbb{R}^n$, $l(x) \in \{1, -1\}$. Each data point is labeled as:

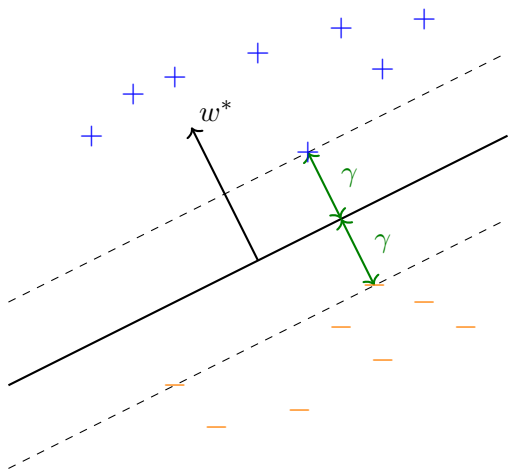
$$l(x) = \text{sign}(\langle w^*, x \rangle) = \text{sign}\left(\sum_{i=1}^{n} w_i^* x_i\right)$$

Here $w^* \in \mathbb{R}^n$ is the unknown normal vector to a separating hyperplane. We assume that $\|w^*\| = 1$ and $\|x\| \leq 1$ for all data points.

Definition 1 (Margin)**.** *The **margin** for w^* is defined as:*

$$\gamma := \min_{x \in D} |\langle w^*, x \rangle|$$

This implies that for any x with label $l(x) = 1$, we have $\langle w^*, x \rangle \geq \gamma$; for any x with label $l(x) = -1$, we have $\langle w^*, x \rangle \leq -\gamma$.



Geometric Interpretation of the Weight Vector

The weight vector w has a fundamental geometric meaning: it is **orthogonal to the separating hyperplane**.

Proposition 2. *The weight vector w is normal to the decision boundary $H = \{x : \langle w, x \rangle = 0\}$.*

Proof. Let $x_1, x_2 \in H$ be any two points on the hyperplane. Then:

$$\langle w, x_1 \rangle = 0 \quad \text{and} \quad \langle w, x_2 \rangle = 0$$

Subtracting: $\langle w, x_1 - x_2 \rangle = 0$. Since $x_1 - x_2$ is an arbitrary vector lying in H, we conclude that $w \perp H$. $\square$

Proposition 3 (Signed Distance to Hyperplane)**.** *The signed distance from a point x to the hyperplane $H = \{z : \langle w, z\rangle = 0\}$ is:*

$$dist(x, H) = \frac{\langle w, x\rangle}{\|w\|}$$

Proof. Let $x_0 \in H$ be any point on the hyperplane, so $\langle w, x_0\rangle = 0$. The signed distance is the projection of $(x - x_0)$ onto the unit normal $\frac{w}{\|w\|}$:

$$\text{dist}(x, H) = \langle x - x_0, \frac{w}{\|w\|}\rangle \tag{1}$$

$$= \frac{\langle x, w\rangle - \langle x_0, w\rangle}{\|w\|} \tag{2}$$

$$= \frac{\langle w, x\rangle - 0}{\|w\|} = \frac{\langle w, x\rangle}{\|w\|} \tag{3}$$

□

The Geometry of Weight Updates

When we update the weight vector $w \leftarrow w + l(x)x$, we are **tilting the hyperplane** toward the misclassified point.

Proposition 4 (Update Improves Classification)**.** *If $(x, l(x))$ is misclassified under w, then after the update $\tilde{w} = w + l(x)x$:*

$$l(x)\langle \tilde{w}, x\rangle > l(x)\langle w, x\rangle$$

Proof. We compute:

$$l(x)\langle \tilde{w}, x\rangle = l(x)\langle w + l(x)x, x\rangle \tag{4}$$

$$= l(x)\langle w, x\rangle + l(x)^2\|x\|^2 \tag{5}$$

$$= l(x)\langle w, x\rangle + \|x\|^2 \tag{6}$$

Since $\|x\|^2 > 0$, the improvement is strictly positive. □

The Perceptron Algorithm

[Perceptron] **Input:** Labeled data $\{(x_i, l(x_i))\}_{i=1}^n$ with $l(x_i) \in \{-1, +1\}$

Output: Weight vector w

1. Start with $w = 0$
2. **for each** input x **do**
   (a) Predict $\text{sign}(\langle w, x\rangle)$
   (b) On a mistake, set $w \leftarrow w + l(x)x$
3. **end for**

Convergence Analysis

Theorem 5 (Perceptron Convergence Bound)**.** *The number of mistakes made by the Perceptron Algorithm on any input sequence is at most* $1/\gamma^2$.

Proof. Consider the potential function $\frac{\langle w, w^* \rangle}{\|w\|}$, starting at 0. Whenever the algorithm makes a mistake, we update $\tilde{w} := w + l(x)x$. We analyze the numerator and denominator separately.

Step 1: Lower bound on the numerator.

$$\begin{aligned} \langle \tilde{w}, w^* \rangle &= \langle w + l(x)x, w^* \rangle && (7) \\ &= \langle w, w^* \rangle + l(x)\langle x, w^* \rangle && (8) \\ &\geq \langle w, w^* \rangle + \gamma && (9) \end{aligned}$$

The last inequality follows from the definition of margin γ: since $l(x) = \text{sign}(\langle w^*, x \rangle)$, we have $l(x)\langle x, w^* \rangle = |\langle x, w^* \rangle| \geq \gamma$.

Step 2: Upper bound on the denominator.

$$\begin{aligned} \|\tilde{w}\|^2 &= \langle \tilde{w}, \tilde{w} \rangle && (10) \\ &= \langle w + l(x)x, w + l(x)x \rangle && (11) \\ &= \langle w, w \rangle + \langle x, x \rangle + 2l(x)\langle w, x \rangle && (12) \end{aligned}$$

Since x is incorrectly predicted by w, we know $l(x) \neq \text{sign}(\langle w, x \rangle)$. This implies that $l(x)\langle w, x \rangle \leq 0$. Also, we assume that $\|x\| \leq 1$. Therefore:

$$\|\tilde{w}\|^2 \leq \langle w, w \rangle + \langle x, x \rangle \leq \|w\|^2 + 1$$

Step 3: Induction after T mistakes.

By induction, after T mistakes:

- Numerator: $\langle \tilde{w}, w^* \rangle \geq \gamma T$
- Denominator: $\|\tilde{w}\| \leq \sqrt{T}$

Step 4: Combining via Cauchy-Schwarz.

By the Cauchy-Schwarz inequality:

$$|\langle \tilde{w}, w^* \rangle| \leq \|\tilde{w}\|\|w^*\| = \|\tilde{w}\|$$

since $\|w^*\| = 1$. This implies that $\frac{\langle \tilde{w}, w^* \rangle}{\|\tilde{w}\|} \leq 1$ at all time steps. Therefore:

$$1 \geq \frac{\langle \tilde{w}, w^* \rangle}{\|\tilde{w}\|} \geq \frac{\gamma T}{\sqrt{T}} = \gamma\sqrt{T}$$

It follows that:

$$\boxed{T \leq \frac{1}{\gamma^2}}$$

□

Lower Bound for the Perceptron Algorithm

Theorem 3.1 shows that the number of mistakes is bounded by $1/\gamma^2$. However, when the margin is tiny, for instance $\gamma \sim 1/2^n$, the number of mistakes can be as large as $2^{O(n)}$, which is not polynomial with respect to the input dimension n.

We construct a dataset and show that for any given order, the Perceptron algorithm needs to see at least $2^{O(n)}$ examples before reaching the truth.

The Hard Instance Construction

For each data point $x^{(i)} \in \mathbb{R}^n$ with i non-zero entries, we construct:

	x	$l(x)$
$x^{(1)}$	$(1, 0, 0, 0, \cdots, 0)$	$+1$
$x^{(2)}$	$(1, -1, 0, 0, \cdots, 0)$	-1
$x^{(3)}$	$(-1, -1, 1, 0, \cdots, 0)$	$+1$
$x^{(4)}$	$(1, 1, 1, -1, 0, \cdots, 0)$	-1
$\vdots$	$\vdots$	$\vdots$
$x^{(i)}$	$((-1)^i, \cdots, (-1)^i, (-1)^{i+1}, 0, \cdots, 0)$	$(-1)^{i+1}$

Lemma 6. *For a unit vector $w^* \in \mathbb{R}^n$ that classifies the data above correctly, its i-th coordinate satisfies $w_i^* \geq 2^{i-1}$.*

Proof. By rescaling w^*, we can assume WLOG that the margin is $\gamma = 1$.

Base case: Since $l(x^{(1)}) = 1$, we have $\langle w^*, x^{(1)} \rangle \geq 1$. This gives $w_1^* \geq 1 = 2^0$.

Inductive step: Similarly, we have $\langle w^*, x^{(2)} \rangle \leq -1$, which gives $w_1^* - w_2^* \leq -1$, and thus $w_2^* \geq w_1^* + 1$.

More generally, for odd i:

$$\langle w^*, x^{(i)} \rangle = -\sum_{j=1}^{i-1} w_j^* + w_i^* \geq 1 \quad \Rightarrow \quad w_i^* \geq \sum_{j=1}^{i-1} w_j^* + 1$$

For even i:

$$\langle w^*, x^{(i)} \rangle = \sum_{j=1}^{i-1} w_j^* - w_i^* \leq -1 \quad \Rightarrow \quad w_i^* \geq \sum_{j=1}^{i-1} w_j^* + 1$$

Therefore, $w_1^* \geq 1$ and for each $i \geq 2$, $w_i^* \geq \sum_{j=1}^{i-1} w_j^* + 1$. By induction, for any $i \in [n]$:

$$w_i^* \geq 2^{i-1}$$

□

Corollary 7 (Lower Bound)**.** *To achieve the correct classifier w^*, the Perceptron algorithm needs at least 2^{n-1} examples with mistakes.*

Proof. In each update step of the Perceptron algorithm, we set $w \leftarrow w + l(x)x$. By the construction of the data above, the absolute value of the change of w is at most one in each coordinate.

By Lemma 4.1, $w_n^* \geq 2^{n-1}$. Therefore, we need at least 2^{n-1} update steps to reach the n-th coordinate of w^*. □

The Modified Perceptron Algorithm

In the previous section, we saw that the Perceptron algorithm cannot be polynomial when the margin is tiny. Here we consider a setting where we only care about data points that are "far" from the margin.

Specifically, given a parameter $\sigma > 0$, we would like to find w such that $\text{sign}(\langle w, x\rangle)$ is the correct label for all x satisfying:

$$|\cos(w, x)| := \frac{|\langle w, x\rangle|}{\|w\|\|x\|} \geq \sigma$$

[Modified Perceptron] **Input:** Labeled data, parameter $\sigma > 0$

Output: Weight vector w

1. Start with $w \leftarrow$ random unit vector
2. **for each** input x **do**
   (a) **if** $l(x) \neq \text{sign}(\langle w, x\rangle)$ **and** $|\cos(w, x)| \geq \sigma$ **then**
   (b) $w \leftarrow w - \langle w, \bar{x}\rangle\bar{x}$, where $\bar{x} = x/\|x\|$
   (c) **end if**
3. **end for**

Theorem 8. *With probability at least* $1/8$*, the number of mistakes made by the Modified Perceptron Algorithm is at most* $\frac{\ln n}{\sigma^2}$.

Proof. Consider the potential function $\cos(w, w^*) = \frac{\langle w, w^*\rangle}{\|w\|\|w^*\|}$.

Step 1: Initialization.

By random initialization, with probability at least $1/8$, we have $\langle w, w^*\rangle \geq 1/\sqrt{n}$.

Step 2: Numerator analysis.

For each update step $w \to \tilde{w}$, since $\text{sign}(\langle w, x\rangle) \neq \text{sign}(\langle w^*, x\rangle)$:

$$\langle \tilde{w}, w^*\rangle = \langle w, w^*\rangle - \langle w, \bar{x}\rangle\langle w^*, \bar{x}\rangle \geq \langle w, w^*\rangle$$

The inequality holds because $\langle w, \bar{x}\rangle$ and $\langle w^*, \bar{x}\rangle$ have opposite signs.

Step 3: Denominator analysis.

Since $|\cos(w, \bar{x})| = \frac{|\langle w, \bar{x}\rangle|}{\|w\|} \geq \sigma$, we have $|\langle w, \bar{x}\rangle| \geq \sigma\|w\|$.

$$\|\tilde{w}\|^2 = \|w\|^2 + \langle w, \bar{x}\rangle^2\|\bar{x}\|^2 - 2\langle w, \bar{x}\rangle^2 \tag{13}$$

$$= \|w\|^2 - \langle w, \bar{x}\rangle^2 \tag{14}$$

$$\leq \|w\|^2(1 - \sigma^2) \tag{15}$$

Step 4: Combining after T steps.

After T steps:

$$1 \geq \cos(w, w^*) \geq \frac{1}{\sqrt{n}} \cdot \frac{1}{(1 - \sigma^2)^{T/2}}$$

Therefore:

$$(1-\sigma^2)^{T/2} \geq \frac{1}{\sqrt{n}}$$

And thus:

$$\left(\frac{1}{1-\sigma^2}\right)^T \leq n$$

Taking logarithms:

$$T \leq \frac{\ln n}{\ln(1/(1-\sigma^2))} \leq \frac{\ln n}{\sigma^2}$$

where in the last inequality we used $1 + x \leq e^x$, which implies:

$$(1-\sigma^2) \leq e^{-\sigma^2} \implies \frac{1}{1-\sigma^2} \geq e^{\sigma^2} \implies \ln\left(\frac{1}{1-\sigma^2}\right) \geq \sigma^2$$

□

Corollary 9 (High-Probability Bound)**.** *By running the Modified Perceptron Algorithm repeatedly for* $\log_{4/3}(1/\delta)$ *rounds, with probability at least* $1-\delta$*, the total number of mistakes is at most:*

$$O\left(\frac{\log n \cdot \log(1/\delta)}{\sigma^2}\right)$$

Proof. By Theorem 5.1, the probability that one round fails is $7/8 \leq 3/4$. The probability that all $\log_{4/3}(1/\delta)$ rounds fail is:

$$(3/4)^{\log_{4/3}(1/\delta)} = \delta$$

So the algorithm succeeds with probability $1-\delta$. The total number of mistakes is at most:

$$\log_{4/3}(1/\delta) \cdot \frac{\ln n}{\sigma^2} = O\left(\frac{\log n \cdot \log(1/\delta)}{\sigma^2}\right)$$

□

From Perceptron to Nonlinear Classifiers

The perceptron finds a linear separator. For nonlinear classification boundaries, we can map data to a higher-dimensional feature space.

Example: Quadratic Boundaries

Consider the label function $l(x) = \text{sign}(\|x - x_0\|^2 - r^2)$, which labels points outside a ball $B(x_0, r)$ as $+1$ and inside as -1. More generally:

$$l(x) = \text{sign}(x^\top Ax - b)$$

For $l(x) = \text{sign}(x_1^2 + x_2^2 - 3x_1x_2)$ with $x \in \mathbb{R}^2$, we can reduce to the linear case by mapping:

$$(x_1, x_2) \mapsto (x_1^2, x_2^2, x_1x_2, x_1, x_2)$$

Then we get a linear classifier in the new coordinates with $w^* = (1, 1, -3, 0, 0)$.

Support Vector Machines

The Perceptron finds *some* separating hyperplane. Support Vector Machines (SVMs) find the **maximum margin** hyperplane.

Hard-Margin SVM

Definition 10 (Maximum Margin Problem)**.**

$$\begin{aligned} \max_{w,b} \quad & \gamma \\ s.t. \quad & y_i(\langle w, x_i \rangle + b) \geq \gamma \quad \forall i \\ & \|w\| = 1 \end{aligned}$$

By rescaling, we can set $\gamma\|w\| = 1$, yielding the equivalent **quadratic program**:

$$\begin{aligned} \min_{w,b} \quad & \frac{1}{2}\|w\|^2 \\ \text{s.t.} \quad & y_i(\langle w, x_i \rangle + b) \geq 1 \quad \forall i \end{aligned}$$

Soft-Margin SVM

For non-separable data, we introduce **slack variables** $\xi_i \geq 0$:

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i \\ \text{s.t.} \quad & y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i \quad \forall i \\ & \xi_i \geq 0 \quad \forall i \end{aligned}$$

The parameter $C > 0$ controls the trade-off between margin maximization and classification error. The term $\|w\|^2$ is the regularization; note that $\gamma_w = \min_x \frac{|\langle w,x \rangle|}{\|w\|}$, so $\|w\|^2 \propto 1/\gamma_w^2$.

Theorem 11 (Perceptron with Slack)**.** *The number of mistakes made by the perceptron is bounded by:*

$$M \leq \frac{1}{\gamma_w^2} + 2\sum_i \zeta_i$$

where ζ_i are the slack variables from the soft-margin SVM.

The Dual Problem

The Lagrangian is:

$$\mathcal{L}(w, b, \xi, \alpha, \mu) = \frac{1}{2}\|w\|^2 + C\sum_i \xi_i - \sum_i \alpha_i[y_i(\langle w, x_i \rangle + b) - 1 + \xi_i] - \sum_i \mu_i \xi_i$$

Taking derivatives with respect to the primal variables and substituting yields the **dual problem**:

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & 0 \leq \alpha_i \leq C \quad \forall i \\ & \sum_{i=1}^{n} \alpha_i y_i = 0 \end{aligned}$$

Key observation: The solution depends on the data only through inner products $\langle x_i, x_j \rangle$.

Definition 12 (Support Vectors)**.** *Points x_i with $\alpha_i > 0$ are called **support vectors**. They are the points closest to (or violating) the decision boundary.*

The optimal weight vector is: $w^* = \sum_{i=1}^n \alpha_i y_i x_i$

Kernel Methods

The Kernel Trick

Since the SVM dual depends only on inner products, we can replace $\langle x_i, x_j \rangle$ with a **kernel function** $K(x_i, x_j)$.

Definition 13 (Kernel Function)**.** *A function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a **valid kernel** if there exists a feature map $\phi : \mathcal{X} \to \mathcal{H}$ (into some Hilbert space) such that:*

$$K(x, z) = \langle \phi(x), \phi(z) \rangle_{\mathcal{H}}$$

Theorem 14 (Mercer's Theorem)**.** *A function K is a valid kernel if and only if it is symmetric and positive semi-definite: for any $\{x_1, \ldots, x_n\} \subset \mathcal{X}$, the Gram matrix $K_{ij} = K(x_i, x_j)$ is positive semi-definite.*

Common Kernels

Kernel	Definition	Feature Space
Linear	$K(x, z) = \langle x, z \rangle$	$\mathbb{R}^d$
Polynomial	$K(x, z) = (\langle x, z \rangle + c)^p$	Monomials up to degree p
RBF/Gaussian	$K(x, z) = \exp(-\gamma \lVert x - z \rVert^2)$	Infinite-dimensional

Why Kernels Work: Implicit High-Dimensional Mapping

[Polynomial Kernel in $\mathbb{R}^2$] For $x = (x_1, x_2)$, $z = (z_1, z_2)$, and $K(x, z) = (\langle x, z \rangle)^2$:

$$K(x, z) = (x_1 z_1 + x_2 z_2)^2 \tag{16}$$

$$= x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2 \tag{17}$$

This equals $\langle \phi(x), \phi(z) \rangle$ where $\phi(x) = (x_1^2, \sqrt{2} x_1 x_2, x_2^2) \in \mathbb{R}^3$.

Kernelized Perceptron

To learn a classifier in feature space $\phi(x)$, we run the perceptron on $\phi(x)$. The weight vector has the form $w = \sum_{i=1}^m \ell(x^{(i)}) \phi(x^{(i)})$, so predictions depend only on inner products:

$$\langle w, \phi(x) \rangle = \sum_{i=1}^m \ell(x^{(i)}) \langle \phi(x^{(i)}), \phi(x) \rangle = \sum_{i=1}^m \ell(x^{(i)}) K(x^{(i)}, x)$$

[Kernelized Perceptron] **Input:** Kernel function K, labeled data

Output: Set M of mistake points

1. Let $M = \emptyset$
2. **for each** input x **do**
   (a) Predict sign $\left(\sum_{y \in M} \ell(y) K(y, x) \right)$

   (b) On a mistake, append x to M

3. **end for**

The algorithm never explicitly computes $\phi(x)$; it only evaluates the kernel K.

Part II: Feedforward Neural Networks

From Linear to Nonlinear Models

The Limitation of Linear Models

Theorem 15 (Composition of Linear Functions is Linear)**.** *If $f(x) = Ax + b$ and $g(z) = Cz + d$, then their composition is linear:*

$$(g \circ f)(x) = C(Ax + b) + d = (CA)x + (Cb + d)$$

Proof. Direct computation. The composition is affine with matrix CA and bias $Cb + d$. □

Corollary 16. *A neural network with only linear activations collapses to a single linear transformation, regardless of depth.*

Stacking layers without nonlinearities provides no additional representational power. A 100-layer linear network is equivalent to a single-layer network.

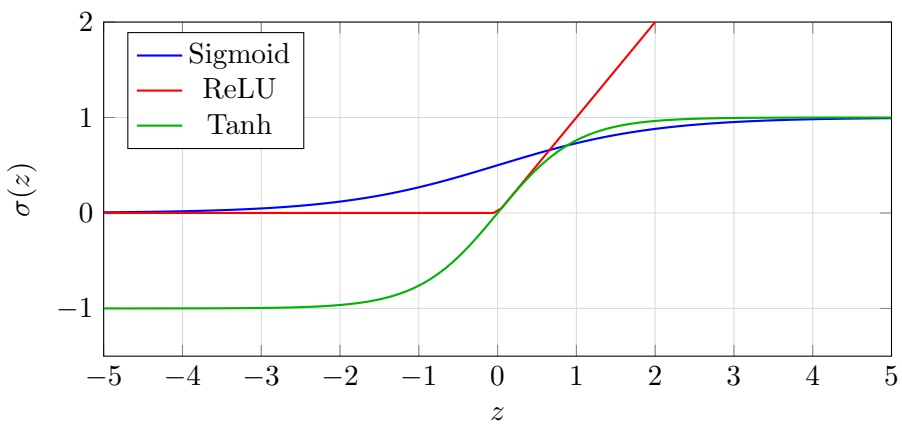
The Need for Nonlinearity

To break the linear limitation, we introduce **activation functions** $\sigma : \mathbb{R} \to \mathbb{R}$ applied element-wise:

$$h^{(l)} = \sigma(W^{(l)} h^{(l-1)} + b^{(l)})$$

Name	Definition	Derivative
Sigmoid	$\sigma(z) = \frac{1}{1+e^{-z}}$	$\sigma(z)(1 - \sigma(z))$
Tanh	$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	$1 - \tanh^2(z)$
ReLU	$\mathrm{ReLU}(z) = \max(0, z)$	$\mathbf{1}_{z>0}$
Leaky ReLU	$\max(\alpha z, z)$	$\alpha \cdot \mathbf{1}_{z \le 0} + \mathbf{1}_{z>0}$

Common Activation Functions



The Universal Approximation Theorem

Statement

Theorem 17 (Universal Approximation). *Let $\sigma : \mathbb{R} \to \mathbb{R}$ be any continuous, non-polynomial activation function. For any continuous function $f : [0,1]^d \to \mathbb{R}$ and any $\varepsilon > 0$, there exists a single-hidden-layer neural network*

$$g(x) = \sum_{j=1}^{N} c_j \sigma(\langle w_j, x\rangle + b_j)$$

such that $\sup_{x\in[0,1]^d} |f(x) - g(x)| < \varepsilon$.

Constructive Proof via ReLU

We give an elementary constructive proof using ReLU activations.

Lemma 18 (Step Function from Two ReLUs). *For any $\epsilon > 0$, define:*

$$h_\epsilon(x) = \frac{1}{\epsilon}\left[ReLU(x-a) - ReLU(x-a-\epsilon)\right]$$

Then $h_\epsilon \to \mathbf{1}_{x\geq a}$ pointwise as $\epsilon \to 0^+$.

Proof. **Case 1:** $x < a$**.** Both arguments are negative, so $h_\epsilon(x) = 0$.

Case 2: $a \leq x < a + \epsilon$**.** Only the first ReLU is active:

$$h_\epsilon(x) = \frac{1}{\epsilon}(x-a) \in [0,1)$$

Case 3: $x \geq a + \epsilon$**.** Both ReLUs are active:

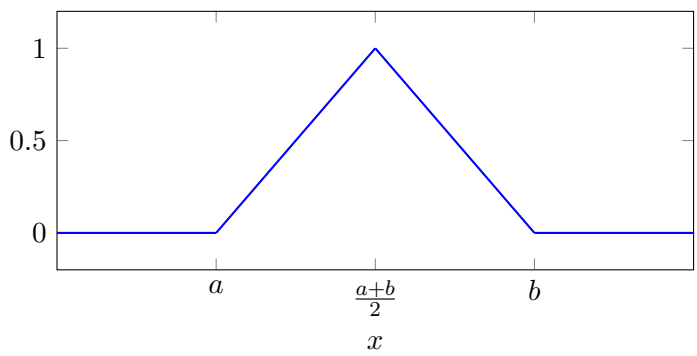
$$h_\epsilon(x) = \frac{1}{\epsilon}[(x-a) - (x-a-\epsilon)] = \frac{\epsilon}{\epsilon} = 1$$

As $\epsilon \to 0^+$, the transition region $[a, a+\epsilon)$ shrinks to a point, giving $\mathbf{1}_{x\geq a}$. □

Lemma 19 (Bump Function). *A "bump" function (or "tent" function) on $[a,b]$ can be constructed:*

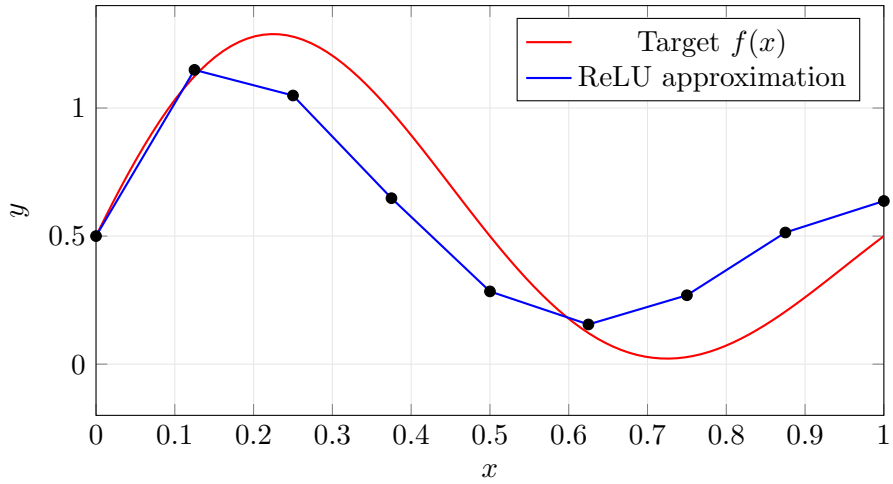
$$bump_{[a,b]}(x) = ReLU(x-a) - 2\cdot ReLU\left(x - \frac{a+b}{2}\right) + ReLU(x-b)$$

Single Bump Function



By scaling and shifting bump functions, we can approximate any continuous function. Each bump is scaled to match the function's value at that region:

Approximating $f(x) = \sin(2\pi x) \cdot e^{-x} + 0.5$ with Scaled Bumps



The approximation improves as we use more bump functions (finer partition). By uniform continuity, for any $\varepsilon > 0$, a sufficiently fine partition yields $\|f - \hat{f}\|_\infty < \varepsilon$.

Theorem 20 (UAT via ReLU, Constructive)**.** *Any continuous $f : [0,1] \to \mathbb{R}$ can be approximated by a finite ReLU network.*

Proof. **Step 1: Partition.** Divide $[0,1]$ into n intervals $[x_{k-1}, x_k]$ of width $1/n$, where $x_k = k/n$.

Step 2: Piecewise linear approximation. Construct $\hat{f}$ by connecting the points $(x_k, f(x_k))$ with line segments. This is a piecewise linear function.

Step 3: ReLU representation. Each piecewise linear function can be written using ReLU "hinges":

$$\hat{f}(x) = f(0) + s_0 x + \sum_{k=1}^{n-1} (s_k - s_{k-1}) \cdot \mathrm{ReLU}(x - x_k)$$

where $s_k = \frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k}$ is the slope of the k-th piece.

Step 4: Approximation error. By uniform continuity of f on $[0,1]$, for any $\varepsilon > 0$, there exists $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$. Choosing $n > 1/\delta$:

$$\|f - \hat{f}\|_\infty \leq \varepsilon$$

Since $\hat{f}$ is a ReLU network with n hidden units, the result follows. □

Formal Analysis via Functional Analysis

Definition 21 (Discriminatory Function)**.** *A function $\sigma : \mathbb{R} \to \mathbb{R}$ is **discriminatory** if for any finite signed Borel measure μ on $[0,1]^d$:*

$$\int_{[0,1]^d} \sigma(\langle w, x \rangle + b)\, d\mu(x) = 0 \quad \forall w \in \mathbb{R}^d, b \in \mathbb{R} \implies \mu = 0$$

Theorem 22 (Cybenko)**.** *If σ is continuous and discriminatory, then the span*

$$\mathcal{S} = span\{\sigma(\langle w, x \rangle + b) : w \in \mathbb{R}^d, b \in \mathbb{R}\}$$

is dense in $C([0,1]^d)$ under the supremum norm.

Proof. Suppose for contradiction that $\overline{\mathcal{S}} \neq C([0,1]^d)$.

Step 1: Apply Hahn-Banach. By the Hahn-Banach theorem, there exists a non-zero bounded linear functional $L : C([0,1]^d) \to \mathbb{R}$ that vanishes on $\mathcal{S}$.

Step 2: Apply Riesz Representation. By the Riesz representation theorem, there exists a non-zero finite signed Borel measure μ on $[0,1]^d$ such that:

$$L(g) = \int_{[0,1]^d} g(x)\, d\mu(x) \quad \forall g \in C([0,1]^d)$$

Step 3: Contradiction. Since L vanishes on $\mathcal{S}$:

$$\int_{[0,1]^d} \sigma(\langle w, x\rangle + b)\, d\mu(x) = 0 \quad \forall w, b$$

But σ is discriminatory, so $\mu = 0$, contradicting the fact that $\mu \neq 0$. □

Lemma 23 (Sigmoidal Functions are Discriminatory)**.** *A function σ is **sigmoidal** if $\lim_{t\to-\infty} \sigma(t) = 0$ and $\lim_{t\to+\infty} \sigma(t) = 1$. Sigmoidal functions are discriminatory.*

Proof. Define $\sigma_{\lambda,\theta}(x) = \sigma(\lambda(\langle w, x\rangle + b) + \theta)$. As $\lambda \to \infty$:

$$\sigma_{\lambda,\theta}(x) \to \begin{cases} 1 & \text{if } \langle w, x\rangle + b > 0 \\ \sigma(\theta) & \text{if } \langle w, x\rangle + b = 0 \\ 0 & \text{if } \langle w, x\rangle + b < 0 \end{cases}$$

By the dominated convergence theorem (since σ is bounded):

$$\int \sigma_{\lambda,\theta}\, d\mu \to \int \mathbf{1}_{H^+_{w,b}}\, d\mu + \sigma(\theta) \cdot \mu(H_{w,b})$$

where $H^+_{w,b} = \{x : \langle w, x\rangle + b > 0\}$ is the open half-space and $H_{w,b}$ is the hyperplane.

If $\int \sigma(\langle w, x\rangle + b)\, d\mu = 0$ for all w, b, then varying θ shows $\mu(H^+_{w,b}) = 0$ for all half-spaces. Since half-spaces generate the Borel σ-algebra, $\mu = 0$. □

Lemma 24 (ReLU is Discriminatory)**.** *The ReLU function $ReLU(z) = \max(0, z)$ is discriminatory.*

Proof. Consider the function:

$$f(x) = \mathrm{ReLU}(\langle w, x\rangle + b_1) - \mathrm{ReLU}(\langle w, x\rangle + b_2)$$

for $b_1 > b_2$. This is a bounded, continuous function. As $\langle w, x\rangle \to +\infty$, $f(x) \to b_1 - b_2$ (constant). As $\langle w, x\rangle \to -\infty$, $f(x) \to 0$.

With appropriate scaling and choice of b_1, b_2, the function $\frac{1}{b_1 - b_2} f(x)$ is sigmoidal. Therefore, if:

$$\int \mathrm{ReLU}(\langle w, x\rangle + b)\, d\mu(x) = 0 \quad \forall w, b$$

then subtracting two such integrals:

$$\int f(x)\, d\mu(x) = 0 \quad \forall w, b_1, b_2$$

Since we can construct sigmoidal functions from differences of ReLUs, and sigmoidal functions are discriminatory, it follows that $\mu = 0$. □

Alternative Proof via Fourier Series

Theorem 25 (Fourier Representation)**.** *Any $f \in L^2([0, 2\pi])$ can be written as:*

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

converging in L^2.

Connection to Neural Networks:

The functions sin and cos can be approximated by neural networks with smooth activations (e.g., tanh, sigmoid). Since any continuous periodic function is a uniform limit of trigonometric polynomials (by Weierstrass approximation), and these can be approximated by neural networks, we obtain UAT.

Alternative Proof via Taylor Series

Theorem 26 (Stone-Weierstrass)**.** *If $\mathcal{A} \subset C(K)$ is an algebra (closed under pointwise multiplication and linear combinations) that:*

1. *Separates points: for $x \neq y$, there exists $f \in \mathcal{A}$ with $f(x) \neq f(y)$*
2. *Vanishes nowhere: for each x, there exists $f \in \mathcal{A}$ with $f(x) \neq 0$*

Then $\overline{\mathcal{A}} = C(K)$.

Application: Neural networks with analytic activations (like sigmoid) can approximate polynomials of any degree (since sigmoid is analytic, products and compositions can approximate any polynomial via Taylor series). By Taylor's theorem, this suffices to approximate any smooth function locally. Patching together local approximations yields global approximation.

Part III: Training Neural Networks

The Training Problem

Loss Functions

Given a neural network $f_\theta : \mathbb{R}^d \to \mathbb{R}^k$ parameterized by weights θ, and training data $\{(x_i, y_i)\}_{i=1}^n$, we minimize the **empirical risk**:

$$\mathcal{L}(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(f_\theta(x_i), y_i)$$

Task	Loss Function	Formula
Regression	Mean Squared Error	$\ell(\hat{y}, y) = \frac{1}{2}\|\hat{y} - y\|^2$
Binary Classification	Cross-Entropy	$\ell(\hat{y}, y) = -y\log(\hat{y}) - (1-y)\log(1-\hat{y})$
Multi-class	Softmax Cross-Entropy	$\ell(\hat{y}, y) = -\log(\text{softmax}(\hat{y})_y)$

Gradient Descent and SGD

Gradient Descent

[Gradient Descent] **Input:** Initial weights θ_0, learning rate η, iterations T

1. **for** $t = 0$ to $T - 1$ **do**
2. $\quad \theta_{t+1} \leftarrow \theta_t - \eta\nabla\mathcal{L}(\theta_t)$
3. **end for**

Per-iteration cost: $O(n)$ gradient computations (one per data point).

Stochastic Gradient Descent

Key Idea: Replace the full gradient with an unbiased estimator using a random sample.

[SGD]

1. **for** $t = 0$ to $T - 1$ **do**
2. $\quad$ Sample $i_t \in \{1, \ldots, n\}$ uniformly at random
3. $\quad \theta_{t+1} \leftarrow \theta_t - \eta_t\nabla\ell(f_{\theta_t}(x_{i_t}), y_{i_t})$
4. **end for**

Proposition 27 (Unbiased Gradient Estimate)**.** *The stochastic gradient is an unbiased estimator:*

$$\mathbb{E}_i[\nabla\ell(f_\theta(x_i), y_i)] = \nabla\mathcal{L}(\theta)$$

Proof.

$$\mathbb{E}_i[\nabla\ell(f_\theta(x_i), y_i)] = \sum_{i=1}^{n}\frac{1}{n}\nabla\ell(f_\theta(x_i), y_i) \tag{18}$$

$$= \nabla\left[\frac{1}{n}\sum_{i=1}^{n}\ell(f_\theta(x_i), y_i)\right] \tag{19}$$

$$= \nabla\mathcal{L}(\theta) \tag{20}$$

□

Convergence Theory

Definition 28 (L-Smoothness)**.** *A differentiable function f is **L-smooth** if its gradient is L-Lipschitz:*

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y$$

Equivalently, for all x, y:

$$f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$$

Definition 29 (μ-Strong Convexity)**.** *A differentiable function f is **μ-strongly convex** if:*

$$f(y) \geq f(x) + \langle \nabla f(x), y - x\rangle + \frac{\mu}{2}\|y - x\|^2 \quad \forall x, y$$

Theorem 30 (GD Convergence, Strongly Convex Case)**.** *If $\mathcal{L}$ is μ-strongly convex and L-smooth, then gradient descent with $\eta = 1/L$ satisfies:*

$$\|\theta_t - \theta^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)^t \|\theta_0 - \theta^*\|^2$$

*The convergence rate is **linear** with rate depending on the condition number $\kappa = L/\mu$.*

Proof. Let $\theta^* = \arg\min \mathcal{L}(\theta)$. By strong convexity applied at θ_t and θ^*:

$$\mathcal{L}(\theta^*) \geq \mathcal{L}(\theta_t) + \langle \nabla \mathcal{L}(\theta_t), \theta^* - \theta_t\rangle + \frac{\mu}{2}\|\theta^* - \theta_t\|^2$$

Since θ^* is optimal, $\mathcal{L}(\theta_t) \geq \mathcal{L}(\theta^*)$. Rearranging:

$$\langle \nabla \mathcal{L}(\theta_t), \theta_t - \theta^*\rangle \geq \frac{\mu}{2}\|\theta_t - \theta^*\|^2$$

By L-smoothness, with $\eta = 1/L$:

$$\|\theta_{t+1} - \theta^*\|^2 = \|\theta_t - \eta\nabla\mathcal{L}(\theta_t) - \theta^*\|^2 \tag{21}$$

$$= \|\theta_t - \theta^*\|^2 - 2\eta\langle \nabla\mathcal{L}(\theta_t), \theta_t - \theta^*\rangle + \eta^2\|\nabla\mathcal{L}(\theta_t)\|^2 \tag{22}$$

$$\leq \|\theta_t - \theta^*\|^2 - 2\eta \cdot \frac{\mu}{2}\|\theta_t - \theta^*\|^2 + \eta^2 L^2\|\theta_t - \theta^*\|^2 \tag{23}$$

Using $\|\nabla\mathcal{L}(\theta_t)\| \leq L\|\theta_t - \theta^*\|$ (by L-smoothness and $\nabla\mathcal{L}(\theta^*) = 0$):

$$\|\theta_{t+1} - \theta^*\|^2 \leq (1 - \eta\mu + \eta^2 L^2)\|\theta_t - \theta^*\|^2$$

With $\eta = 1/L$:

$$\|\theta_{t+1} - \theta^*\|^2 \leq \left(1 - \frac{\mu}{L}\right)\|\theta_t - \theta^*\|^2$$

□

Theorem 31 (SGD Convergence, Non-Convex Case)**.** *Under L-smoothness and bounded variance $\mathbb{E}[\|\nabla\ell_i - \nabla\mathcal{L}\|^2] \leq \sigma^2$, SGD with constant step size $\eta = \frac{1}{L\sqrt{T}}$ satisfies:*

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla\mathcal{L}(\theta_t)\|^2] = O\left(\frac{L(\mathcal{L}(\theta_0) - \mathcal{L}^*)}{\sqrt{T}} + \frac{\sigma}{\sqrt{T}}\right)$$

Proof. By L-smoothness:

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) + \langle \nabla \mathcal{L}(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{L}{2}\|\theta_{t+1} - \theta_t\|^2$$

Substituting $\theta_{t+1} - \theta_t = -\eta g_t$ where g_t is the stochastic gradient:

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \eta\langle \nabla \mathcal{L}(\theta_t), g_t \rangle + \frac{L\eta^2}{2}\|g_t\|^2$$

Taking expectations conditioned on θ_t, using $\mathbb{E}[g_t|\theta_t] = \nabla \mathcal{L}(\theta_t)$ and $\mathbb{E}[\|g_t\|^2|\theta_t] = \|\nabla \mathcal{L}(\theta_t)\|^2 + \sigma^2$:

$$\mathbb{E}[\mathcal{L}(\theta_{t+1})|\theta_t] \leq \mathcal{L}(\theta_t) - \eta\|\nabla \mathcal{L}(\theta_t)\|^2 + \frac{L\eta^2}{2}(\|\nabla \mathcal{L}(\theta_t)\|^2 + \sigma^2)$$

Rearranging:

$$\|\nabla \mathcal{L}(\theta_t)\|^2 \leq \frac{1}{\eta(1 - L\eta/2)}\left[\mathcal{L}(\theta_t) - \mathbb{E}[\mathcal{L}(\theta_{t+1})|\theta_t]\right] + \frac{L\eta\sigma^2}{2(1 - L\eta/2)}$$

Taking full expectations and summing from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] \leq \frac{\mathcal{L}(\theta_0) - \mathcal{L}^*}{\eta(1 - L\eta/2)} + \frac{TL\eta\sigma^2}{2(1 - L\eta/2)}$$

Setting $\eta = \frac{1}{L\sqrt{T}}$ (which satisfies $L\eta/2 = \frac{1}{2\sqrt{T}} < 1/2$ for $T \geq 1$):

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[\|\nabla \mathcal{L}(\theta_t)\|^2] = O\left(\frac{L(\mathcal{L}(\theta_0) - \mathcal{L}^*)}{\sqrt{T}} + \frac{\sigma}{\sqrt{T}}\right)$$

□

Computing Gradients: Forward vs. Backward

Neural Networks as Computational Graphs

A feedforward neural network $f : \mathbb{R}^d \to \mathbb{R}$ can be viewed as a **directed acyclic graph** (DAG) where:

- Input nodes: $x_1, \ldots, x_d$
- Intermediate nodes: $v = \sigma(u)$ where $u = \sum_k w_k v_k$ is a weighted sum of parent nodes
- Output node: f

The goal: Compute $\frac{\partial f}{\partial w}$ for every weight w in the network.

The Chain Rule for Computational Graphs

Consider a node v in the network. Let u be the weighted sum before the activation:

$$u = \theta_1 t_1 + \theta_2 t_2 + \cdots + \theta_n t_n$$

$$v = \sigma(u)$$

By the chain rule:

$$\frac{\partial f}{\partial \theta_1} = \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial \theta_1}$$

- $\frac{\partial v}{\partial u} = \sigma'(u)$ - depends only on the activation function
- $\frac{\partial u}{\partial \theta_1} = t_1$ - the input to this weight
- $\frac{\partial f}{\partial v}$ - **this is the challenge**: how does changing v affect the final output?

The Multivariable Chain Rule

When node v has multiple downstream nodes $w_1, \ldots, w_m$:

$$\frac{\partial f}{\partial v} = \sum_{j=1}^{m} \frac{\partial f}{\partial w_j} \cdot \frac{\partial w_j}{\partial v}$$

Forward Mode Differentiation

Idea: For each weight θ, propagate $\frac{\partial v}{\partial \theta}$ forward through the graph.

Algorithm (Forward Mode):

1. Fix a weight θ_{ij} in layer l
2. Initialize $\frac{\partial v}{\partial \theta_{ij}} = 0$ for all nodes v before the weight
3. At the node where θ_{ij} appears: $\frac{\partial v_i}{\partial \theta_{ij}} = \sigma'(u_i) \cdot v_j^{(l-1)}$
4. Propagate forward through all downstream nodes using the chain rule
5. At the output, we have $\frac{\partial f}{\partial \theta_{ij}}$

Cost: $O(E)$ per weight, where E = number of edges. For W weights: $O(WE)$ total.

The Inefficiency of Forward Mode

Key Insight (Gundersen): In forward mode, we compute *the same terms repeatedly.*

Consider computing $\frac{\partial f}{\partial \theta_i}$ and $\frac{\partial f}{\partial \theta_k}$ for two weights in the same layer:

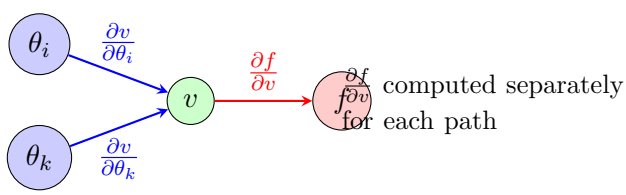
$$\frac{\partial f}{\partial \theta_i} = \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial \theta_i}$$

$$\frac{\partial f}{\partial \theta_k} = \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} \cdot \frac{\partial u}{\partial \theta_k}$$

The terms $\frac{\partial f}{\partial v}$ and $\frac{\partial v}{\partial u}$ are **repeated** but computed separately in forward mode for each weight.

The fundamental problem: At node v, we cannot compute $\frac{\partial f}{\partial v}$ (which depends on downstream nodes) without visiting all downstream nodes. In forward mode, we must do this separately for each weight.

Forward Mode:



Backpropagation: The Efficient Solution

Key Insight: If we compute $\frac{\partial f}{\partial v}$ for all nodes v in a *backward* pass, we can share these computations across all weights.

The multivariable chain rule tells us:

$$\frac{\partial f}{\partial v} = \sum_{j:w_j \text{ child of } v} \frac{\partial f}{\partial w_j} \cdot \frac{\partial w_j}{\partial v}$$

If we already have $\frac{\partial f}{\partial w_j}$ from the backward pass, we can compute $\frac{\partial f}{\partial v}$ using only **local** information.

[Backpropagation] **Forward Pass:**

1. Compute all activations $v^{(l)}$ layer by layer
2. Store intermediate values $(u^{(l)}, v^{(l)})$ for the backward pass

Backward Pass:

1. Initialize $\delta^{(L)} := \frac{\partial \mathcal{L}}{\partial v^{(L)}}$ at the output layer
2. **for** $l = L - 1$ down to 1 **do**
   (a) Compute $\delta^{(l)} = (W^{(l+1)})^\top \delta^{(l+1)} \odot \sigma'(u^{(l)})$
   (b) Compute $\frac{\partial \mathcal{L}}{\partial W^{(l)}} = \delta^{(l)} (v^{(l-1)})^\top$
   (c) Compute $\frac{\partial \mathcal{L}}{\partial b^{(l)}} = \delta^{(l)}$
3. **end for**

Here $\delta^{(l)} = \frac{\partial \mathcal{L}}{\partial u^{(l)}}$ is the "error signal" at layer l, and $\odot$ denotes element-wise multiplication.

Theorem 32 (Backprop Complexity)**.** *Backpropagation computes all partial derivatives in $O(E)$ time, where E is the number of edges (weights) in the network.*

Proof. Each edge is visited exactly once in the forward pass (to compute activations) and exactly once in the backward pass (to compute gradients). Each visit requires $O(1)$ operations. Therefore, total time is $O(E)$. □

Comparison:

Method	**Time Complexity**
Numerical differentiation	$O(WE)$ (finite differences)
Forward mode autodiff	$O(WE)$
Backpropagation	$O(E)$

The Vanishing and Exploding Gradient Problem

The Vanishing Gradient Problem

Consider a deep network with L layers and sigmoid activations. The gradient at layer l involves:

$$\frac{\partial \mathcal{L}}{\partial W^{(l)}} \propto \prod_{k=l}^{L} \sigma'(u^{(k)}) \cdot W^{(k)}$$

The Issue: For sigmoid, $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

Proposition 33. *The sigmoid derivative satisfies $\sigma'(z) \leq \frac{1}{4}$ for all $z \in \mathbb{R}$, with equality at $z = 0$.*

Proof. $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Let $s = \sigma(z) \in (0, 1)$. Then $\sigma'(z) = s(1 - s)$.

By AM-GM: $s(1 - s) \leq \left(\frac{s+(1-s)}{2}\right)^2 = \frac{1}{4}$, with equality when $s = 1/2$, i.e., $z = 0$. □

Theorem 34 (Vanishing Gradients)**.** *For a network with sigmoid activations and L layers, if $\|W^{(k)}\| \leq 1$ for all k:*

$$\left\| \frac{\partial \mathcal{L}}{\partial W^{(1)}} \right\| = O\left(\frac{1}{4^{L-1}}\right)$$

Gradients vanish exponentially with depth.

The Exploding Gradient Problem

Conversely, if $\|W^{(k)}\| > 4$, gradients can **explode** exponentially:

$$\left\| \frac{\partial \mathcal{L}}{\partial W^{(1)}} \right\| = O\left(\lambda^{L-1}\right) \quad \text{for } \lambda > 1$$

Solutions

1. ReLU Activation:

$$\sigma'(z) = \begin{cases} 1 & z > 0 \\ 0 & z \leq 0 \end{cases}$$

Gradients are either 0 or 1; no shrinking occurs in the active region.

2. Careful Initialization (Xavier/He):

- **Xavier (Glorot):** $\mathrm{Var}(w) = \frac{2}{n_{\text{in}} + n_{\text{out}}}$ (for tanh/sigmoid)
- **He:** $\mathrm{Var}(w) = \frac{2}{n_{\text{in}}}$ (for ReLU)

The idea is to keep the variance of activations constant across layers.

3. Batch Normalization:

Normalize activations within each mini-batch to have mean 0 and variance 1:

$$\hat{z} = \frac{z - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

This keeps activations in the "active" region of the activation function.

4. Residual Connections (ResNets):

$$h^{(l+1)} = h^{(l)} + F(h^{(l)})$$

Gradients can flow directly through the identity path, avoiding the vanishing gradient problem.

Summary

The Progression of Ideas

Model	Key Insight	Limitation
Perceptron	Linear separation via hyperplane	Only linearly separable data
SVM	Maximum margin + kernel trick	Fixed feature space
Neural Network	Nonlinear activation $\to$ UAT	Training is non-convex
Deep Networks	Depth gives expressiveness	Vanishing/exploding gradients

Key Theorems

1. **Perceptron Convergence:** At most $1/\gamma^2$ mistakes
2. **Perceptron Lower Bound:** $\Omega(2^{n-1})$ mistakes possible
3. **Modified Perceptron:** $O(\log n/\sigma^2)$ mistakes with high probability
4. **Universal Approximation:** Single hidden layer suffices for any continuous function
5. **Backprop Efficiency:** $O(E)$ vs $O(WE)$ for forward mode

Practical Guidelines

1. Use ReLU (or variants) for hidden layers
2. Use proper initialization (He for ReLU, Xavier for tanh/sigmoid)
3. Consider batch normalization for deep networks
4. Monitor gradient norms during training
5. Use residual connections for very deep networks (> 20 layers)